TPDN90G2 V. 130A RELEASE NOTE

■ **Adopt new version for all new tape out and all designs immediately** ☐ **Adopt new version for all designs, both on going and new start** ☐ **Adopt new version for new design start** ☐ **No action required**

1. LIBRARY INFORMATION

Library Name: *TPDN90G2*

Library Version: 130

Release Note Version: A

Library Type: Standard I/O

Technology: *90NM LOGIC GENERAL PURPOSE 1P9M SALICIDE 1.0V/2.5V (CLN90G)*

Library Status: *Silicon Proven*

Design Rule: T-N90-LO-DR-001 v1.3

Spice model: T-N90-LO-SP-002 v1.4

DRC runset: T-N90-LO-DR-001-C1 v1.3a

LVS runset: T-N90-CL-LS-001-C1 v1.2b

LPE runset: T-N90-LO-SP-002-B1 v1.4b

Cell number: 31 (18 I/O cells + 1 corner +6 fillers + 6 bonding pads)

2. UPDATE HISTORY

<table>
<tr><th>Date</th><th>Version</th><th>Changes</th><th>Customers action recommended</th></tr>
<tr><td>12/22/04</td><td>130A</td><td>1. Update the library to comply with design rule V1.3</td><td>Adopt new version for all new tape out and all designs immediately</td></tr>
<tr><td rowspan="6">1/2/04</td><td rowspan="6">110A</td><td>1. Silicon proven.</td><td rowspan="6">Adopt new version for all new tape out and all designs immediately</td></tr>
<tr><td>2. Update design rule to V1.1</td></tr>
<tr><td>3. Adopted to V1.0 Spice/LVS & re-char. (Using BSIM4 model & including LOD effect)</td></tr>
<tr><td>4. Remove Star-DC (sdc) kit</td></tr>
<tr><td>5. Add CeltIC CDB (ctc) kit</td></tr>
<tr><td>6. Include input enable (IE) function<table><tr><th>PAD</th><th>IE</th><th>C</th></tr><tr><td>x</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table></td></tr>
</table>

<table>
<tr><td>1/2/04</td><td>110A</td><td>7. Change cell name & configuration
<table>
<tr><td>V100B:</td><td>V110A:</td></tr>
<tr><td>PDCLCDG_25</td><td>PDC0204CDG_25</td></tr>
<tr><td>PDCMCDG_25</td><td>PDC0816CDG_25</td></tr>
<tr><td>PDCHCDG_25</td><td>PDC1224CDG_25</td></tr>
<tr><td>PDSLCDG_25</td><td>PDS0204CDG_25</td></tr>
<tr><td>PDSMCDG_25</td><td>PDS0816CDG_25</td></tr>
<tr><td>PDSHCDG_25</td><td>PDS1224CDG_25</td></tr>
<tr><td>PRCMCDG_25</td><td>PRC0816CDG_25</td></tr>
<tr><td>PRCHCDG_25</td><td>PRC1224CDG_25</td></tr>
<tr><td>PRSMCDG_25</td><td>PRS0816CDG_25</td></tr>
<tr><td>PRSHCDG_25</td><td>PRS1224CDG_25</td></tr>
<tr><td>PVDD1CDG_25</td><td>PVDD1CDG_25</td></tr>
<tr><td>PVDD2CDG_25</td><td>PVDD2CDG_25</td></tr>
<tr><td>PVDD2POC_25</td><td>PVDD2POC_25</td></tr>
<tr><td>PVSS1CDG_25</td><td>PVSS1CDG_25</td></tr>
<tr><td>PVSS2CDG_25</td><td>PVSS2CDG_25</td></tr>
<tr><td>PVSS3CDG_25</td><td>PVSS3CDG_25</td></tr>
<tr><td>PXOE1CDG_25</td><td>PXOE1CDG_25</td></tr>
<tr><td>PXOE2CDG_25</td><td>PXOE2CDG_25</td></tr>
<tr><td>PCORNER_25</td><td>PCORNERG_25</td></tr>
<tr><td>PFEED0_005_25</td><td>PFEED0005G_25</td></tr>
<tr><td>PFEED0_01_25</td><td>PFEED001G_25</td></tr>
<tr><td>PFEED0_1_25</td><td>PFEED01G_25</td></tr>
<tr><td>PFEED0_5_25</td><td>*</td></tr>
<tr><td>PFEED1_25</td><td>PFEED1G_25</td></tr>
<tr><td>PFEED5_25</td><td>PFEED5G_25</td></tr>
<tr><td>*</td><td>PFEED10G_25</td></tr>
<tr><td>PFEED35_25</td><td>PFEED35G_25</td></tr>
<tr><td>PADIZ40_25</td><td>PAD40N_25</td></tr>
<tr><td>PADOZ40_25</td><td>PAD40G_25</td></tr>
<tr><td>PADIZ45_25</td><td>PAD45N_25</td></tr>
<tr><td>PADOZ45_25</td><td>PAD45G_25</td></tr>
<tr><td>PADLZ60_25</td><td>PAD60G_25</td></tr>
<tr><td>PADLZ85_25</td><td>PAD85G_25</td></tr>
</table></td><td></td></tr>
<tr><td>07/09/03</td><td>100B</td><td>1.Update LVS netlist of PVDD1CDG_25
2. Update PAD cell layout to fix metal floating issue
3. New database structure, re-generate lpe.</td><td>Adopt new version for all new tape out and all designs immediately</td></tr>
<tr><td>04/15/03</td><td>100A</td><td>1. Update design rule to v0.3
2. Update spice model to v0.3
3. Revise PVDD1CDG to use thick oxide ESD trigger circuit due to leakage</td><td>Adopt new version for all new tape out and all designs immediately</td></tr>
</table>

		4. Remove VDD-VDDPST diode in PVDD1CDG	
		5. Add tie high resistor in PVDD2POC	
		6. Change OD2 to OD25	
		7. Add dummy MT/OD/PO blockage layer	
		8. Re-layout metal above double guard ring	
		9. Improve driving capability	
		10. Remove Poly & Pimp under PAD	
		11. Add POC PMOS metal option to reduce leakage	
		12. Reduce OD&OD25 density	
07/03/02	020A	First Release	Adopt new version for all new tape out and all designs immediately

3. DESIGN KIT VERSION DEPENDENCY TABLE

The following table shows the valid combinations of design-kit versions

rln	*gds*	*gdf*	*spi*	*apf*	*apt*	*sef*	*set*	*syn*	*vlg*	*vit*	*sdc*	*mdt*	*doc*	*lpe*	*ibs*	*ctc*
130a	130a	-	130a	130a	130a	130a	-	130a	130a	130a	-	130a	130a	130a	130a	130a
110a	110a	-	110a	110a	110a	110a	-	110a	110a	110a	-	110a	110a	110a	-	110a
100b	100b	-	100b	100b	100b	100b	-	100a	100a	100a	100a	100a	100a	100b	-	-
100a	100a	-	100a	100a	100a	100a	-	100a	100a	100a	100a	100a	100a	100a	-	-
020a	020a	-	020a	020a	020a	020a	-	020a	020a	020a	020a	020a	020a	020a	-	-

rln = release note
gds = GDSII layout view
gdf = GDSII phantom view
spi = Spice/lvs netlist
apf = Apollo FRAM (phantom) view
apt = Apollo FRAM and CEL (layout) view
sef = Silicon Ensemble LEF (phantom) view
set = Silicon Ensemble LEF and Virtuso (layout) view

syn = Synopsys
vlg = Verilog
vit = Vital (VHDL)
sdc = Star-DC
mdt = Mentor DFT (Fastscan, DFT advisor)
doc = Documents (Datasheets)
lpe = Layout Parasitic Extracted Spice Netlist
ibs = ibis model file
ctc = Celtic CDB view

4. CHARACTERIZATION CONDITIONS

TYPICAL	**BEST**	**WORST**	**LOW-TEMP**
VDDIO=2.5Volt VDDCORE=1.0Volt Temp=25C Process= NMOS: Typical PMOS: Typical	VDDIO=2.75Volt VDDCORE=1.1Volt Temp=0C Process= NMOS: Fast PMOS: Fast	VDDIO=2.25Volt VDDCORE=0.9Volt Temp=125C Process= NMOS: Slow PMOS: Slow	VDDIO=2.75Volt VDDCORE=1.1Volt Temp= -40C Process= NMOS: Fast PMOS: Fast

5. KNOWN PROBLEMS AND LIMITATIONS

a. Please do not use smaller filler cells to fill large I/O spacing. DRC errors will be reported if small fillers are inserted to large spacing.

b. TLF files were derived from Synopsys .lib files; however, the CT-GEN requires a non-decreasing trend in delay table when loading increases. Therefore, the TLF files in this library had been fixed to meet this requirement. Users can generate his/her own TLF files directly from the Synopsys .lib files using the Cadence utility "syn2tlf" if needed.

c. All types of resistor are recognized as user defined devices (X-card) in new layout versus schematic check (LVS, V1.0). Please include the file 'source_added' provided by LVS runset into the customers' final source netlist when performing LVS check.

6. SPECIAL NOTES

a. Please use V04.40-PO31 or newer version of celtic for cdb kits.

b. The library contains a cell named PVDD2POC_25 for POC25 (power on control) signal generation. **It is MANDATORY to implement "one and only one" PVDD2POC_25 cell per separated digital I/O domain.** This can be achieved by replacing "one" PVDD2CDG_25 cell with "one" PVDD2POC_25 cell per each separated digital I/O domain. Forgetting to do so would result in a floating POC25 rail, which would affect the functions of I/O buffers, and would in turn jeopardize the whole chip performance.

c. Spice netlist tpdn90g2_1_2.spi is for separated GND (VSS and VSSPST25) application, and tpdn90g2_3.spi is suitable for same core and I/O GND application.

7. LIBRARY CONTENTS

Cell List: (18 I/O cells + 1 corner + 6 fillers + 6 bonding pads)

I/O Cells:	**Corner Cell:**	**Pad Cells:**	**Feeders:**
pdc1224cdg_25	pcornerg_25	pad40n_25	pfeed0005g_25
pdc0816cdg_25		pad40g_25	pfeed05g_25
pdc0204cdg_25		pad45n_25	pfeed1g_25
pds1224cdg_25		pad45g_25	pfeed5g_25
pds0816cdg_25		pad60g_25	pfeed10g_25
pds0204cdg_25		pad85g_25	pfeed20g_25
prc1224cdg_25			
prc0816cdg_25			
prs1224cdg_25			
prs0816cdg_25			
pvdd1cdg_25			
pvdd2cdg_25			
pvdd2poc_25			
pvss1cdg_25			
pvss2cdg_25			
pvss3cdg_25			
pxoe1cdg_25			
pxoe2cdg_25			

8. TAPE-OUT LAYER INFORMATION

a. Please closely follow TSMC 90nm CMOS Logic 1P9M Salicide 1.0V/2.5V Masking Layers and Bias document.

b. Please refer to Chapter of "Library Integration Notes" of TSMC I/O Application Note for general tape-out information.

c. It is required to tape out with layers listed in the following table. For the rest of layers necessary for tape out, please refer to TSMC 90nm CMOS Logic 1P9M Salicide 1.0V/2.5V Masking Layers and Bias document.

(Continued on the following page)

Tape Out Layer Name	Layer Number			
	6ML Tape Out	7ML Tape Out	8ML Tape Out	9ML Tape Out
NWELL	3	3	3	3
OD	6	6	6	6
NT_N	11	11	11	11
OD_25	41	41	41	41
POLYG	17	17	17	17
PIMP	25	25	25	25
NIMP	26	26	26	26
RPO	29	29	29	29
CONT	30	30	30	30
Metal 1	31	31	31	31
Via 1	51	51	51	51
Metal 2	32	32	32	32
Via 2	52	52	52	52
Metal 3	33	33	33	33
Via 3	53	53	53	53
Metal 4	34	34	34	34
Via 4	54	54	54	54
Metal 5	35	35	35	35
Via 5	55	55	55	55
Metal 6	36	36	36	36
Via 6	-	56	56	56
Metal 7	-	37	37	37
Via 7	-	-	57	57
Metal 8	-	-	38	38
Via 8	-	-	-	58
Metal 9	-	-	-	39
Passivation (CB)	43	43	43	43
RHDMY	117	117	117	117
ESD3DMY	147	147	147	147